DRAFT REPORT

MASTER THESIS IN INTERACTION DESIGN

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11 MAY 2020

fluid logic



ABSTRACT

Within the field of education tools like Arduino are used to teach students relevant skills, attitude and knowledge around technology. Based on interviews with experts, observations in the classroom, reflection with the teachers and a workshop with Arduino this thesis sketches a future vision that re-designs the tools to be more collaborative and fluid. Since these tools are often used in group contexts and group work is an important part of the learning; could we design tools that not just make it possible but invite for collaborative interactions. Also how could the tools allow for sketching and iterating more fluidly so reflection, action and re-action cycles could be smaller and allow for more exploration and learning.

The goal of this report is to inpsire you; to invite you into my project and brain to allow you to doubt my directions while holding your hand throughout this journey. If you would ever want to reach out; to talk about this topic or if things are unclear feel free to contact me; through whatever way communication happens at that moment in the (virtual) reality.

TABLE OF CONTENT

p4 In **the background** I will invite you into why I think this topic is relevant for society, for interaction design and for me as a designer. And I will give you a overview of my understanding of the field.

p16 In the **rage on the keyboard**; I'll show how norms are engrained in tools by seeing the world throught the lens of the keyboard.

p18 After that I will guide you through my **approach** that describes the methods, tools and interventions within the context that I did during my design process.

p20 In the **results** I will describe what these activities in my approach lead to.

p34 In the **analysis** I wil start showing how the foundations in the results lead to design principles and directions for my final design

p40 In the **design** section I will go into depth on specific features of the design and why they are formed the way they are.

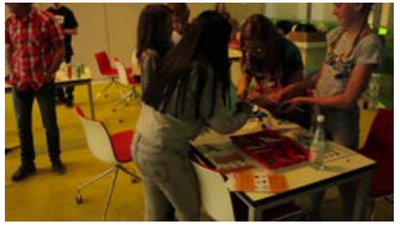
p52 In the **frequently asked questions** I'll answer questions from teachers, students, researchers that I referred to.

p60 In the **reflection** I will reflect on what I learned.

p62 In the **references** I will guide you through some material that I've been using throughout my process.

p63 Last but not least, **the appendix** gives a richer overview of my time planning, collaborations and includes some documentation.





BACKGROUND

The scope of this project is to improve the software tools when learning physical computing systems within the context of high-school education, for students aged 14-16 year old. In the research phase I've done fieldwork in different types of high-school education; after the research phase I've narrowed down on design based learning class.

Personal background

I have a history of teaching and developing course material for Arduino since 2011; In workshops at the ICTdag I was giving teachers in IT, physics and related fields an introduction into Arduino and give examples of how it could be implemented in education. The focus of these workshops would to be get the teachers hands dirty as soon as possible. I used the contacts that I've build up over time to contact and visit schools where they are currently teaching Arduino in the classroom. Besides this I've been teaching a course in problem solving with LEGO mindstorms for 6 lessons to elementary and high-school students. In my internships at LEGO product technology, Arduino education and Microsoft Research I've been looking into learning experiences as well. Based on these experiences I thought it would be interesting to apply ethnographic methods to understand what is (actually) happening in the classrooms at the moment; for this reason an important part of this thesis is in the research both through talking about the observations, talking about the tools and actually being in the class not as a teacher; but observer to get a different perspective on the whole field, and use this as an input for the human-centred design approach (in industry there might be more factors like sales, marketing and technology-constraints narrowing the solution space, before it ends up in the design teams). The design work in this thesis is an example of how the research could be translated into the form of a product; but throughout the thesis I've been suggesting and proposing smaller changes to existing systems as well.

21st century skills

Our current society is full of technology. To make sure that the next generation is prepared to use and steer this development we need to educate them with tools that focus on the long term; while giving them understanding of what it is they interact with.

Tools define our imagination

The tools we use to create and use are defining our imagination. By changing the aesthetics of these tools we could allow a broader group of people to envision different futures. Arduino has started with this mission, bringing the technology that was only usable by engineers in labs to artists, designers and education. In this way the not only the visions from the priviledged people in research labs can be materialised, but everyone can join and contribute to the development of new technologies.

"The tools we use have a profound (and devious!) influence on our thinking habits, and, therefore, on our thinking abilities" (Dijkstra 1982).

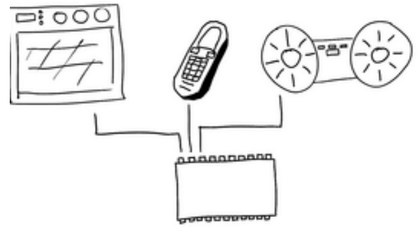
"We don't always have ideas and then express them in the medium. We have ideas with the medium." (diSessa 2000, p. 116)

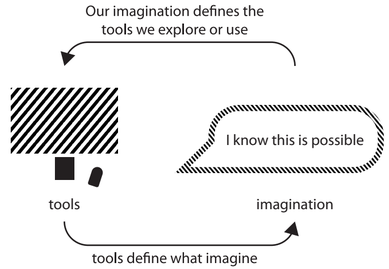
Consumerism to activism

I would like the next generation of 'users' to be able to move from being passive consumers to more active creators of the future. To achieve this children need to be able to explore what is possible; see the bigger map and not just follow the path that is pre-defined by the design of the product.

In the PhD thesis of David quartiles this was coined as the toolkit vs toolbox approach. A toolkit might help to guide the user to achieve the goal of the toolkit; a toolbox has more of an open-end, it might be used for purposes that the designer, educator or producer of the toolbox didn't intend.

Although through standardisation in education we tend to give students





"**Kits have a certain intentionality**, as when dealing with preconceived designs. It is not that the recipients cannot change the kit's fate, but success cannot be guaranteed unless it is in the prescribed manner." - (Quartiles, 2018).



more of the toolkit type of experiences than toolbox type of experiences. F.e. in maths education we give students a tool, but not to create; but just to repeat; even in the more arts type of education, such a language learning; we tend to focus on filling in the gaps type of exercises or teach students grammar; instead of letting them write essays.

We should consider if we want students to learn through these types of products; that reinforce the norms of our current consumerist society. In (Redström, J, 2006) Johan Redström reflected on the fact that we might be designing users instead of designing objects that empower users.

Action centered design

This type of design is different than normal human-centred design, since there is no clear problem that needs a solution. The goal is to open up a soluation space for others to explore. We need to think more about it like designing a pencil, the goal is not to only fill in checkboxes on an exam, it is also to create a drawing for your friends, this is a very different context but still we are able to use the same tool for this situation. Programming should have the same type of aestetics as a pencil in that sence. (D. Norman, 2005)

Aesthetics of interaction

I will use the aesthetics of interaction as a way to envision different ways of interacting with technology. By bringing examples of the type of aesthetics early on into the project that I want to achieve I'll try to break with current paradigms and let others share their ideal way of interacting with making technology.

Contextual

Students should learn technology in contexts, preferably close to the source where it is being used. Making tools that only serve education feels like you're not teaching the students the 'real' thing. By not hiding everything behind a box, but opening up what is inside it creates a better understanding of what it is.

Stop analysing start doing

Arduino is a cool way to explain 'embodied' logic to students; the focus should be on the action and exploration not on endlessly analysing. The shorter the feedback loops the better the understanding is. They should feel encouraged to explore; instead of being paralysed by the complexity.

Coding is social

When people hear of coding it is usually not the most social thing you can think of. This is sad, because this leads to self-taught programmers that never learned how to communicate code. While this is (especially looking forward) the most important part of coding. How can you create a common knowledge of what is happening. How can coding become more authoring (a tool to communicate something) instead of engineering (a tool to communicate to a computer).
Many of the hard-skills like programming will probably be different in the future. For this reason I think that the education should be more focussed on the authoring than the 'programming'.

This tool to communicate to each other about what is happening and why also forces reflection; it forces students to think through the code. Besides that it is important for understanding.

Users pick an (online) example and tweak it to do what they want in a different context. This process of re-using code is perhaps more important than actually writing it; but the way in which it is communicated than becomes more important.

Teachers

It is hard for teachers to keep up with the developments around technology education. This is one of the reasons that students and teachers are moving to a more self-directed way of learning; where the information that the students learn is not limited to the teacher's knowledge. My goal in this project is to figure out how I could improve this way of self-directed learning.

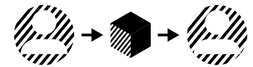
consumerism (user is just using the artifacts)



making/coding (user is making the artifacts)



authoring (user is making the artifacts
as a tool to communicate with others)



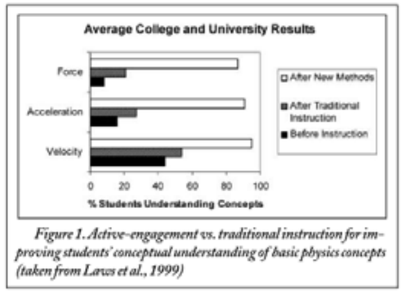


Figure 1. Active-engagement vs. traditional instruction for improving students' conceptual understanding of basic physics concepts (taken from Laws et al., 1999)

This chart shows how effective short active collaborative interventions are in teaching lectures for explaining physics concepts in engineering class.

Since the context of the Arduino ecosystem is a blend between digital skills (programming, analytical thinking, problem-solving and designing) and physical learning (soldering, understanding and exploring hardware and 'trying-out') it would be great to design touchpoints for the communities, schools and perhaps industry to work together to make the learning experience inclusive and accessible to the beginning users.
This will lead to students that are better in searching for information, better in working together (and sharing their work for others to build upon), better in problem-solving and more aware of the possibilities and potential risks of technology in their everyday life.

Learning theories

Active cooperative & Collaborative learning

The review paper by (M. Prince, 2004) also shows **a significant benefit of active learning in groups**, the paper shows that small group interventions already have a huge benefit on the learning outcomes (see chart). It increases the engagement by students measured in increase of people continuing their studies (22%) and improves the retention of traditionally underrepresented groups. Although individual responsibility should not be absent. I think it's insane that most engineering lectures are still taught the traditional way for 50 minute lectures; where people remember 70% of the information presented in the first 10 minutes and 20% of the information in the last 10 minutes. If you're reading this as a teacher giving lectures; note down; give students every 15 minutes 2 minutes to discuss their notes.

The myth of learning styles

There has been a history of 'learning style' theories that have influenced our view on education. The idea of learning styles is that students have a preferred method (visual, auditory, physical, social...) that would perform best to learn a topic. This theory is criticised and **proven to be faulty** in various experiments. (W. Daniel, 2015) The idea that learning styles are personal attributes to the students isn't the case; the styles are mostly related to the content; so if the content is well represented in a particular manner than that would help all students, having multiple representations can help bridging understanding.

Friction and memory retention

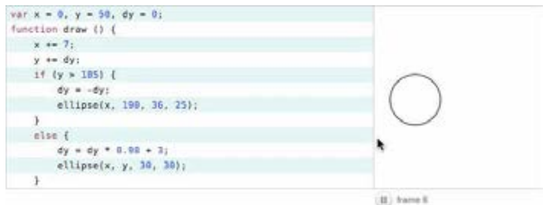
In 'Make it stick' (P. Brown, 2014), the main message is that **learning should be effortful**; it is best to space practise in the example of a text book, it would be best to learn not each concept after each other; but learn them in parallel. This seems like a painful way to learn; but that is exactly their point learning that is easily divided into nice blocks will not stick, since what you're learning is the book, not the content and concepts behind it, if you have to put effort into 'finding' the concept in your brain it will build pathways to this knowledge.

Block programming

Programming for beginners is a minefield, there has been efforts into making programming more understandable by enclosing text into boxes. (Weintrop, D., & Wilensky, U. ,2017), which helps the understanding, exploration and enjoyment. But students still respond as positively to the question "Programming is hard", since what they did in class **is not seen as the 'programming'** that the question asked.

Learnable programming

The article with this title by (B. Victor, 2012) has been highly influential for me during this thesis, in his essay he pointed out five things that the environment should allow the learner to **read the vocabulary, follow the flow, see the state, create by reacting, create by abstracting**. This essay is definitely worth a read if you're interested in how we can change programming environments to be more transparent and understandable. In my final concept I will refer back to these concepts.

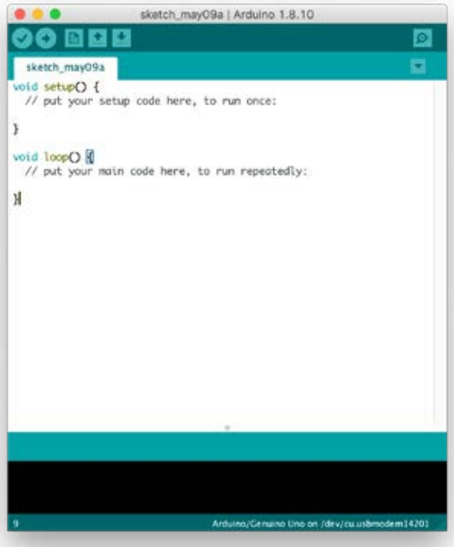


Arduino IDE

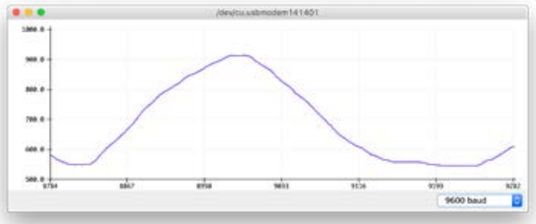
The Arduino IDE is the tools that makes Arduino what it is; it is a notepad like interface similar to Processing that allows you to write code, compile this code and upload it to the board.

Debugging

Besides these functions Arduino IDE also allows for debugging; textual commands that are send from the code could be represented in two ways; textual or if the commands contain numbers on a visual chart. This last functionality is added by (H. Pohl, 2015), and teachers have been appreciating this to demonstrate and explain concepts.







Sketch

The sketch is the name for code in Arduino. It uses cpp with on top a layer of functions that interact with the hardware.

Errors

This area underneath is reverved for errors from the different parts of the assemblage, by my observations in the classroom these errors were often too cryptic and ignored by the students.

History of Arduino

Arduino is an open-source hardware and software solution developed in Interaction Design Institute Ivrea (IDII) and was based on the thesis work by Hernando Barragán on Wiring. The Arduino IDE was build on top of the processing environment. The goal of Arduino was to create low-cost and easy way for students (in interaction design) to create devices that could interact with their environment using sensors and actuators.

Arduino did a great job in packaging the tooling; originally when you want to set up a programming enviroment this might take a day; which is effort that could have been put somewhere else. Since the Arduino IDE is an assemblage of all the nessecary parts to make, compile and upload a piece of code to the Arduino it was a huge improvement over collecting these tools together yourself.

Arduino education

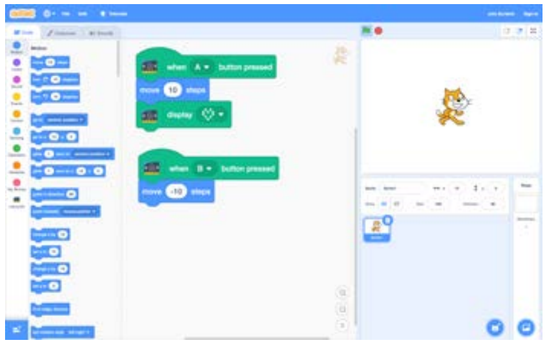
Arduino Education is a brach of Arduino that focusses on making curriculums, platforms and tools for students in middle-school till university to learn by using Arduino. Some of these products focus more on the coding side of things; other more on data analysing.

Besides developing the material Arduino education also gives support on distance for teachers in the form of webinars and organises events CTC Faires for the students to present and see other's projects.





Makecode by microsoft, https://makecode.microbit.org/#editor



Scratch by MIT media lab, https://scratch.mit.edu/projects/editor/

Tools in the context of education

In the past years there have been a lot of kickstarter crowdfunding campaigns around the field of education and physical computing. Besides that there have been successful online platforms that teach programming, this thesis is not an extensive overview of what has been done; but here is a short summary of the current platforms mentioned by high-school teachers.

Micro-bit

Micro-bit is a programmable system on a PCB that has on-board components together with the possibility to programme these boards with various development environments:

- Makecode by microsoft
- Python by the Python software foundation
- Scratch by MIT media lab

This 'solves' the problem of having to focus on tree things at the same time (coding, electronics and building); one could merely focus on the programming; and even simulate the behaviour in a browser; there is the possibility to extend the capabilities beyond the on-board component by the digital and analog in/output ports on the PCB, altough this is not common practise.

LEGO

The LEGO group also has a brach for education; to teach programming and physical computing. This branch of products ranges from pre-school (Spike prime) to high-school (Mindstorms) and is used in various types of activities such as the first LEGO league, where students in groups compete to automate a task using LEGO based hardware.

In LEGO products the messiness of the reality is abstracted away by putting components in bricks that are connected through a standardised set of connectors that are easy to connect and control by interfaces on phones, laptops or tablets.

Besides creating the products LEGO creates a network of teachers, and makes curriculums for their educational products.

Adafruit

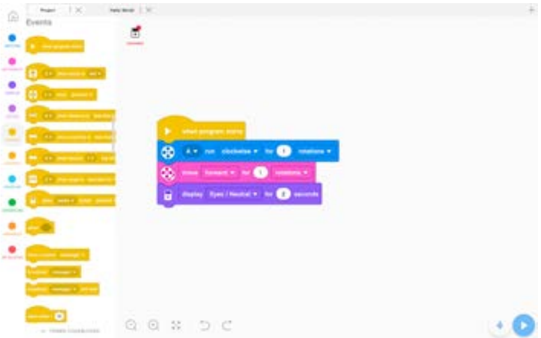
Adafruit is a company that focusses on selling hardware to the maker community mostly; but started to branch out to education as well, mostly by working together with code.org by making circuitplayground; which consists of two different versions; the Classic version and the Express version. The last version works with makecode, CircuitPython, Arduino and code.org. Adafruit doesn't seem to sell full solutions (kits, platform and support) to teachers but gives the oppertunity for the teachers to mix and match their own curriculum based on the documentation and tutorials that exist.

Raspberry pi

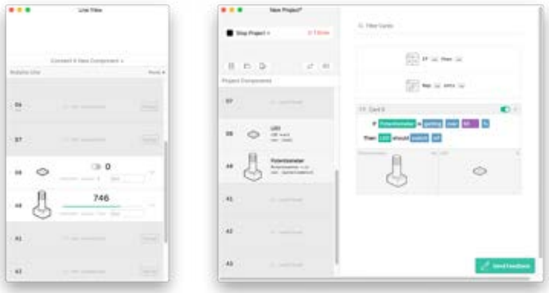
Raspberry pi is a hardware platform that is based on a full operating system (usually linux) and is originally focussed on education; but mostly adopted by the DIY community because of the price and flexibility. Since there is no clear one-path on how to use it in the field of education for teachers it is still hard to implement it in their curriculum; another reason why it is not widely used in education is the setup time and effort. In most use cases the raspberry pi is the operating system on which the coding happens. Things like the pi-top emerged from this.

Blockdots

Blockdots is not made for education, but still worth a mention. It is a tool for desingers to prototype interactive prototypes. It connects to an Arduino board and is an interactive way of defining mapping between in and output ports.

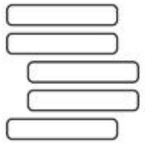


LEGO mindstorm EV3 for education



Blockdots interface.

Procedural metaphor



c++
python
...

Programming paradigms

Here I will give a short overview on what are the current and previous paradigms and understandings around programming.

Procedural programming

Procedural programming comes from the foundation of imperative programming where you tell the computer what to do at a certain moment; in the past commands like goto were used to make loop structures, currently if, while and for loops replace this system of goto, since it allows for scoping of variables. This is still the most popular metaphor, since it is close to what is actually happening in the current single core hardware.

Functional programming

Functional programming comes from languages like LISP and functional features are now implemented by popular languages like javascript and c++. The idea of a purely functional language is that there are no side effects; like in mathematics the function takes arguments and returns a value, not influencing or reading information that is outside these two (sets) of values. Functional programming has the benefit that it might run better on parallel computers.

Object oriented programming

Object oriented programming makes sure that variables are encapsulated in objects; so they have their own scope and only interact with other objects using public functions. This helps to make sure that it becomes more tidy and 'boxed' and is also the structure in which libraries function in Arduino.

Logic languages

Logic languages focus on defining the outcome and not the process to get there; especially in planning and scheduling software this helps to abstract the problem and focus on setting constraints for the ideal solution, and let the computer figure out how to get there.

Programming by intent

This idea that users can describe their intention is an interesting view on what programming is; could we explain our intention by giving examples? In the interface of Excel this type of programming synthesis is implemented that allows end-users to write (mostly text based manipulations) that would otherwise be done by regular expressions by giving one or two examples and let excel figure out what they intended. (S. Gulwani) Also software like Eureqa (R. Ehrenberg, 2012) showed that software can generalise from specific examples; since it created the equation for the double pendulum.

Representations by flow

The idea of flow based programming has been around for a long time as well and is mostly powerful in interactive programming with data flows, the idea is that the messages flow from one node to another node. Node-red by IBM is one example of this type of representation of code.

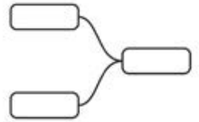
Representations by table

This idea of using the data itself as representation for the structure of the code has been around as well. Many people would argue that Excel is not 'real' programming, but that is probably why it is so widely used and powerful. (B. Kankuzi, 2017) In one of the classrooms I went they we're using Arduino in combination with excel, which worked really good, there was not that much focus on the tooling, most of the focus was on the actual learning; in that case learning physics.

Representations by timeline

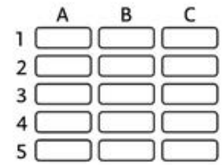
The idea of using a timeline to visualise (inter)actions is known mostly in the field of visual animation software such as flash, director and after effects. Also in the field of physical computing CEE by nonolith labs was a way to both send and monitor analog values through the timeline metaphor. Teachers also used timeline's on blackboards, used the Serial plotter to explain sensors and make problems by the students understandable through writing a little timeline. Arduino projects also have a tendency to have a time-based property in them, in the current procedural metaphor students sometimes have trouble understanding why two programmes can't be combined (see Ubi's interview in results).

Flow metaphor



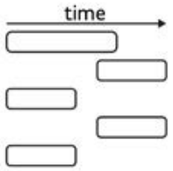
Node-red
Pure data
maxMSP

Spreadsheet metaphor

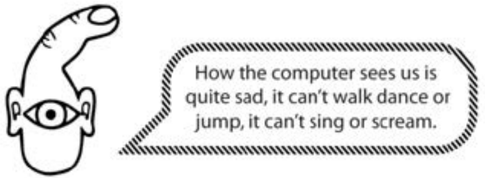


Excel
Google spreadsheets
...

Timeline metaphor



AS (Adobe Flash)
Director
...



From Physical Computing: Sensing and Controlling the Physical World with Computers, (Tom Igoe, 2004, Page 14)



One piece of text written in Digital body language (https://awards.ixda.org/entry/2019/digital-body-language/)

RAGE ON THE KEYBOARD

I'm writing this part in the sun on a bank in the area in from of my student dorm in Sweden; sitting in the sun; and listening to myself from yesterday night; ranting about the keyboard. Yesterday we had a lecture from Pernilla Glaser about storytelling; with the main message to be "anarchistic" in the way we tell stories. So now I'm sitting here raging on the keyboard; about the keyboard.

The keyboard is fascinating; it allowed people to share stories, write them down, communicate over time; over distance from their mind into a different shape. Since the invention of paper and pen, we have been writing down thoughts in letters and word. But the keyboard did something different; the keyboard quantified the writing of the pen into a standardised way of writing, a standardised way of seeing, a standardised way of understanding and a standardised way of creating the world. It's impact is not only in the act of typing; but everywhere around us in the world; in science that we teach students, the signs on the street and the way we communicate with each other.

Let's start by taking it apart; a keyboard has keys; with on these keys a visual representation of a symbol; letters that we use to write stories. The words that appear on the screen than represent real things; but we tend to loose the connection the reality.

And we lost a lot of visibility of the intent; writing a letter on the keyboard always has the same 'academic gloss' over it. It doesn't matter is you spend 10 minutes or 10 hours on it. It will loose the dept, it will loose the amount of times you repeatedly tried to write something, crossed things out. Digital Body Language from Malmö university beautifully shows the fact that we loose important information about the process.

To come to programming where we have on top of the already existing 'academic gloss' of the keyboard we have another type of 'gloss' that is in the programming language.

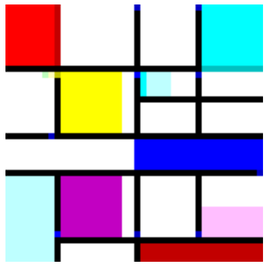
The lack of real experimentation to make it different or make different ways of understanding possible is concerning. People think through the keyboard; through the devices that they use around them to express themselves. If we don't consider alternative ways of thinking and expressing. We end up in a world where everything is made around efficiency. It is as if we only invented a paint-roller to do programming; but didn't invent the paintbrush yet. We did re-invent the handle of the paint roller, but didn't really re-invent the tool itself. We need to reconsider what programming is; perhaps by starting with specific instead of general.

This lack of re-thinking paradigms is beautifully represented by the keyboard; the QUERTY layout that we inherited from the typewriter is still there; while it forces people into a way of working that is not 'optimal' ergonomically for the people; but for the technology that doesn't exist anymore. This is not only true for the keyboard, but is almost a property of technology.

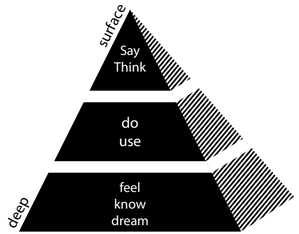
Another norm that is highly visible through looking at the keyboard is the fact that it is made for individuals; the idea that an individual is expressing thoughts, this can be seen through the body that the keyboard lives in or next to LAPtop, PERSONAL computer, iMac; the software that runs on the computer and the way we organise our digital platforms. The only software I know that focusses on multiple users on one keyboard is a game called Zatacka (or Achtung die kurve); and when playing it; it exposes the awkwardness of multiple people behind one keyboard.

You can see how this 'caging' of the human mind takes place in how we comment in code; you can see that people can adjust to systems that are limiting them; but the question is do we want this? Do we want people to express their art in just the whitespace around the code? Is it because or despite the tools that we reached the point where we are now? And are we now moving in the right direction?

It is not necessary to stick to the old paradigms, we don't have to see the computer as a typewriter; because it can be more; and it our duty as interaction designers to make it 'more'.



Piet a programming language that works by colors this program above writes piet to the console (T. Schoch)



(Visser, F. S. 2005).

APPROACH

Here is an overview of the methods that I've been using throughout the design process. This pyramid helped to define what type of research something is; and to make sure that I'm consciously choosing the different ways of engaging throughout the process.

In the top layer is the say and think layer; this is where interviews, questionnaires and other more traditional ways of engaging with the context come into the picture.

The second layer is the do and use layer; in this layer it is not about what people say they do; but what people actually do; this can be discovered by observations or monitoring, through f.e. step-counters etc.

The feel know dream layer relates to what futures people admire, build and dream of. This layer is best to discover through the practise of co-creation. By making tools in which participants can join the design process they can express this layer of information. This is an important layer since it is related to body storming and using different ways of engaging than only talking.

During the process I tried to make use of video as much as possible, to keep people updated on the process, to have a clear overview of the documenting process and to reflect-on-action. I booked time in my calendar every Friday to do this type of reflection through video.

Research phase

In the research phase I explored the use of Arduino in various formal education classes with students aged 13–14 years in the Netherlands. With this research I aimed to explore the main challenges and opportunities that arise when using Arduino in formal education. To explore these different ethnographic methods were used: 1) participant observation; supporting the teacher(s) by helping groups 2) fly on the wall observation; I was in the back of the classroom to observe. After each session of (participant) observation, I had a 3) recorded open interview with the teacher to reflect on and explore certain aspects during the class.

I approached schools and teachers that I know from giving workshops around Arduino in education.

Besides this observation and reflection in formal education I also visited an artist that is creating a different way to program, experts in the field of computer science and education and other teachers through online interviews.

Itterations/ design phase

My ambition was to do at least one full itteration with user-testing at a school, but because of the covid-19 situation it was harder to contact, travel and do this user-involvement during the design phase of the project. So instead of user-testing in reality; I've been presenting and getting feedback as a way to validate the directions. I've developed several itterations/ directions which by working on them also made me better in understanding the problem and solution space. Thes experiments or itterations I used to map out the solution space, to make consious decisions on what the focus should be; and what the focus shouldn't be.

RESULTS

Interviews

Ubi de Feo

Ubi is a hybrid designer and very curious human being working at Arduino.

According to Ubi de Feo it is very common, that users in the beginning stage are not being able to change or understand the code written by others. Take the two simple examples: one blinking an LED and the other using a button as input. They both work fine on their own, but when you try to blend them you have to have more understanding about than just the two sketches. Then you have to talk about time-slots, talk about multitasking, parallel computing etc.

He thinks we should steer away from the trend of abstracting everything.

Education should open a new thinking pathway; it should give the students a way of thinking around the problem that is not by the book; it should allow students to learn beyond the book.

Education should give students a framework or context that makes the students curious about something, my goal is to get you so interested that you want to learn how to talk to computers. This is not done by just delivering topics but by creating a maker environment where it's more dialogue of learning and transferring knowledge.

According to Ubi the paradigm of a timeline "doesn't seem to bad". Every time that he has to address a problem that doesn't make sense to him; He takes paper and write down what happens with t0, t1, t2, t3.

In the couse from 0 to C, Ubi used concepts that we're related to computers by acting, play out the experience, "Everyone likes a bit of theatre."

Alvin Arthur

Kinaesthetics designer, educator & creative director



I visisted Alvin Arthur in his studio in Eindhoven; where we talked about embodyment in programming and he showed me his latest prototypes.

Alvin Arthur is currently working on bodyHTML a setup with the Kinect to create web-based pages, where div's and images are being created by a pair of dancing programmers; one programmer is the builder (more choreographic way of thinking, structure) and one the tweaker (the more freestyler that defines the assets, images and text; where it is like scrolling trough content/images). The tweaker can adjust the content, the builder is creating the structure; what goes in which div, what order.

The goal is to make a collaborative environment instead of competitive environments; where students need to work together to achieve tasks.

He focusses on web-pages since everyone knows what a webpage is; everyone can relate to it. It's not going to replace keyboards (because they are too advanced), but according to Alvin programming shouldn't only be for keyboard people or mouse people. Since kids get frustrated if they don't move. He thinks we shouldn't make them sit to think, but make them move to explore.

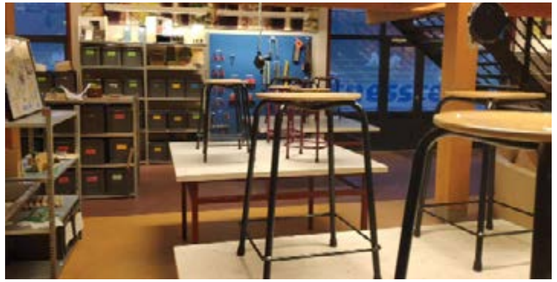
Frank Hagenaars

Teacher Onderzoek&Ontwikkelen (research and development en T&T Technologie en Toepassing (Technology and application), owner webshop and trainer for teachers

I try to make students help each other; if one group of two is really understanding it I refer them to a group of students that is struggling more. After taking this course; 10 students didn't really get it to be honest; these students you would never get excited for Arduino).



The setup in the physics class



A video of the design based learning class

Field study

This study explored the use of Arduino in various formal education classes with students aged 13–14 years in the Netherlands. The main insights that emerged from the research are balancing acts for the teachers, students and (learning) tools within this context. These balancing acts are 1) roles vs togetherness in group work, 2) ‘learning to’ vs ‘learning by’ concerning the friction of the tools, 3) teachers focussing on giving technical support vs supporting active learning and learning materials giving ownership vs guidance.

Methods

The research aims to explore the main challenges and opportunities that arise when using Arduino in formal education. To explore these different ethnographic methods were used: 1) participant observation; supporting the teacher(s) by helping groups 2) fly on the wall observation; the researcher was located in the back of the classroom to observe. After each session of (participant) observation, the researcher had a 3) recorded open interview with the teacher to reflect on and explore certain aspects during the class.

Context

This research is done in physics class, technology class and design-based learning (DBL) class. In physics class, the students were experimenting and using Arduino as a tool for collecting the results in groups of 4 students. The goal was to make students familiar with the hardware and learn how analog-to-digital conversion works. Arduino was pre-programmed to act as a keyboard to type in the values of a temperature sensor. In technology class, the students were following a sequence of steps in groups of 2 students. The goal was to learn the basic instructions to manipulate outputs. In the DBL class, the students were working in groups of 4 students to create a scale model of an escape room with an Arduino based challenge. The goal of using Arduino in this context was creating awareness of its possibilities.

All teachers pointed out they were using Arduino so that students knew it’s existence and capabilities so students could choose to use it in future student-lead science or design projects. The fact that there is a huge community sharing Arduino projects and extensions (outside the field of education) was a major reason for choosing this platform.

Findings

Roles vs togetherness in group work

In all the researched scenarios students were working in groups since teachers pointed out that otherwise, it wouldn’t be manageable. Teachers said that they aimed at changing the roles after each activity but in reality, the students that started doing the programming stayed behind the laptop and seemed usually the students with most experience and confidence to use the Arduino or technology in general.

“I try to make them switch roles as much as possible. In every exercise, they should switch roles. In reality, this doesn’t really happen, since not everyone is at the same exercise, I can’t do this with a call for attention.”- High-school teacher technology class (translated from Dutch)

Within the DBL class, it was interesting to see how the students working on the Arduino part were disconnected from the materiality of the rest of the group work. Other students in the groups were (drilling, painting, cutting) and having more togetherness by iteratively sharing results. This might come from the properties of the tools and the lack of opportunities for togetherness provided by the tool. This might reinforce the stereotypical programmer as a lone person; focussed and disengaged looking at a screen.

Practical issues

Here are some pracitcal notes from the observations, that I put as a side note for the people interested.

Losing the code because you saved them locally on the laptop; Arduino defaults to save the sketches on the laptop; because of this reason, a lot of people lost the code when they opened another laptop this class.

Finding the right Serial port; this is something that they memorize after several times having this problem, but not really supporting the explore-ability of trial and error.

Lack of computational motor-skills (triple-click to select line) or knowledge about shortcuts (CTRL-V, CTRL-C) and for example how to write curly-brackets made it hard to write; copy and paste and created errors.

The ; (semicolon) at the end of each instruction is usually forgotten

The == is usually written as a = (if not directly copy-pasted)

Merging sketches was a big problem; what needs to be kept and what needs to be replaced. This led to multiple definitions of the same variable.

The difference between the language, variables and syntax is hard to understand. What is defined by the author (of the copied code)? To be able to make adjustments to the code this is really important.

The position in the code; where to put code. Since for students, it feels like an arbitrary block of text it is hard for them to know where to put something.

The friction of the tools; Learning by vs Learning to

Friction is an important tool for learning [2] so the place and way in which this friction is experienced should be well considered. The place of the friction should be on the activities that are in line with the learning outcomes. In the case of high-school, this is usually on skills that can be applied to other problems “learning by using Arduino” instead of focussing only on “learning to use Arduino”.

Examples of the Arduino specific friction could be selecting the serial port, the invisible nature of what code is doing, error messages and function in jargon and students’ second language that are hard to decipher or spell right.

This friction is also due to the lack of diversity of representations of code. Code relies heavily on symbolic visual understanding; this is true for the error messages, the code and the learning material itself.

Examples of transferable learning or friction could be separating problems into subproblems, understanding logic, understanding mapping and communicating behaviour or scenarios.

Technical support vs supporting active learning

The activity of the teacher was usually to walk around the classroom and respond to questions from students. These questions were mostly related to the troubleshooting that they were doing with the hardware and software. For the teachers, it was sometimes tricky to balance teaching (a.k.a. supporting active learning) and giving technical support because of several aspects: sometimes the teacher didn’t have an overview of the problem, time pressure, and spreading time and effort over the different students or groups.

“It’s a very stressful and terrifying experience when as a teacher something doesn’t work, and you don’t know why. You only have so much time; every time I teach an Arduino workshop … At some point, you have to give up and leave some people behind, so it’s a lose-lose situation.”—Workshop teacher & Employee Arduino Education

Ownership vs guidance

The learning material is balancing between having open-ended exercises and guided instructions. For students, the lack of ownership of their creations mismatched their learning expectations.

“We don’t learn that much from copying code. If I’m here anyway I better learn anything”—Student technology class (translated from Dutch).

Teachers said that some guided instructions were needed to teach the basics, only in the physics class students didn’t have any previous introduction and could use Arduino as a measurement tool after a less than 5-minute intro, (the focus was not on Arduino since it was pre-programmed but on the experiment).

Discussion

These balances are being used to map out some areas of improvements: 1) move the friction to match transferable learning outcomes, such as understanding mapping, working in a group and communicating intent. 2) Make the tools more ‘visible’ for teachers and students to see what is happening, this might lead to more active learning by the student and helps the teacher to support this process of active learning more without acting as technical support to map out the problem. 3) Make tools more open for group interactions, right now due to the tools (laptop with IDE that mostly uses a keyboard as input) it is hard for students to work on the coding/ defining the behaviour part together; while in the observations Arduino was always used in groups. The tools should reflect this and invite for group interactions. 4) Make the art of defining behaviour more explorative. currently ‘playing around’ is hard with the existing tools, due to the friction and rigid structure that coding has.



Experiments

In this section I will guide you trough my experience of using other existing tools and re-formatting them to create explain and explore a different idea. In this case more visual-based tools are used to define behaviour of more physical based interactions in the real world.

The paragraph below is written for the design and DIY community on the website project hub; which was a double edged sword; it helped me understanding how sharing in the community works and it helped me explaining my concept for a broader audience. This is more aimed at the design community which is usually quite familiar with graphic design tools (adobe suite) and bridge this to the physical computing field.
There are two benefits to using already existing tools:

People are already familiar with them, if they are well known in the community
It is an easy (low-cost) way to imagine different ways of defining behaviour, than writing code

Below there are three experimental setups described with each different possibilities, limitations and powers.

PIXEL TO ARDUINO

This video shows how the light of the NeoPixels is influenced by the position of the cursor. In this way it is an easy way to prototype different light behaviour in Arduino.

The goal is to see how defening behaviour could be extended from the current paradigms that are around programming. This example really shows the power of direct control to increase understanding of the thing the students are creating.

Published on projecthub as “From pixel to neopixel“

PHOTOSHOP INTERACTION MATRIX

This prototype explores using photoshop as a way to prototype an ‘interaction matrix‘. You can than use two inputs to influence 3 different outputs; such as 2 axis servo motors or an RGB color.

The goal is to see how different representations of behaviour could be sketched; while keeping the dynamic behaviour.

It creates perhaps more ‘visible’ code and makes artistic expressions easier than logic. It works good for light; but it is harder for other things like movement.

I wrote a Medium article titled “Art of mapping“ about this explorations

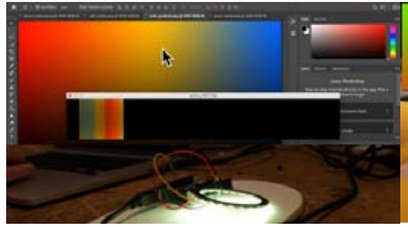
KEYNOTE AS CODE

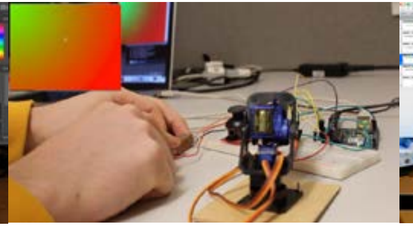
This prototype explores using keynote as a way to prototype a scenario; a part of the presentation is reserved for influencing the output; the other part of the presentation can give context in the form of text and images.

The goal is to see if and how and if it would make sense to merge storytelling with programming to increase the togetherness of code.

By focussing on already existing and learned skills we could allow for more people to be involved.

It might make telling and defining the story together easier. Doesn’t explain or show the complexity. Doesn’t increase understanding of the hardware; probably more the opposite.







Workshop

This study describes the findings from doing a workshop at Arduino; where the goal was to share the research and gather insights from these experts in the field of Arduino and education. This was done in tree different ways. Sharing our own experiences and bringing my research into this. Acting out scenario's based on the classroom-research; to get an active understanding of the context and something to point to. And ideating based on metaphors and inspiration from other fields (sports, music , dance, games and other hobbies).

Preparation

"The goal of the session will be to sketch the future of learning Arduino with the focus on the interactions with the tools (Arduino, laptops…) and people (excited students, passive students and teachers) and how we can imagine better ways of interacting that will increase the learning, fun, teamwork and change the stigmas around programming." - Information from email send to participants.

Before the session I asked the participants to bring stories from childhood education, what are moments that are still with you at the moment. How they would explain blink to a 10-year old student, get the participants to think about representations; how mental models could be visualised or explained. What are three main reasons to use Arduino in education; what are the core values. And to bring three main problems that occur when using Arduino in Education.

One part of the workshop was around sharing the insights from the research done in the schools, I shared this in different ways; using the three positive and negative points based on the research; that we could than map together with theirs. I brought stories and scenario's from the research for acting out the context, and ideas of how to solve the main problems with the IDE right now.

Clustering the insights



HURDLES

It is hard to get started it has a steep learning path programming is now a pre-requisite for starting to explore it.



TEACHING

Students give up, it is terrifying a a teacher, lack of teamwork while useing Arduino



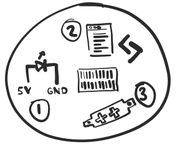
COMPLEXITY

Lots of different components, it is not clear what compatibility



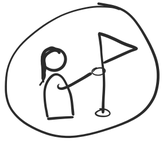
INSTABILITY

Setup needs guidance, instabilities in uploading and connecting, these are types of friction that don't increase understanding



LEARNING MULTIPLE THINGS

You have to learn 3 things at the same time (electronics, programming and mechanical building), difficult troubleshooting



OWNERSHIP

Ownership as a main reason for motivation, feels like magic, making means ownership



BEYOND EDUCATION

Using real tools, real code and electronics (that are used outside the school context), Allows students to go deep, Arduino is portable (bring home/to context), Changes the way you think of the world, make your ideas reality

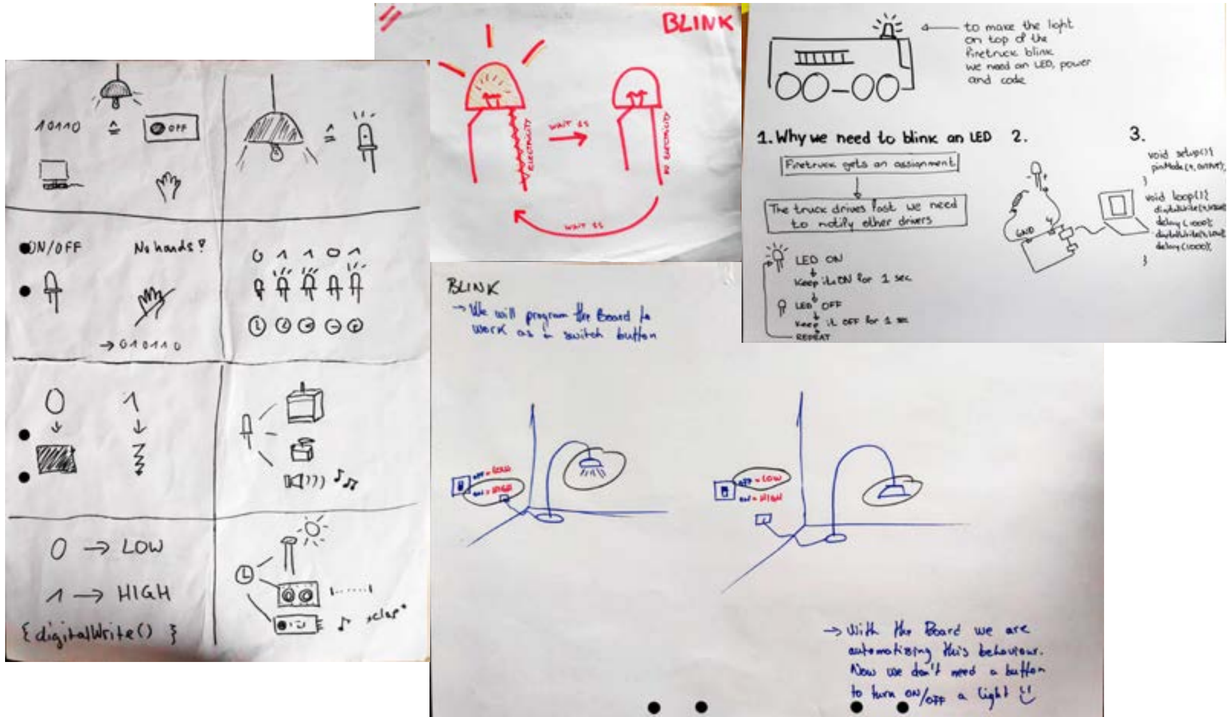


THE PHILOSOPHY

Sharing and community, no ceiling to what is possible, you learn to make stuff instead of only consume.

Representation of blink

Everyone brought a description on how to describe blink to a ten-year old. One powerful way to represent it is the metaphors. One employee showed an firetruck as a way to make people understand what a blinking light is; one used the living room light as an example.



Acting

Acting out the scenario's was a great way to understand the dynamics; as one participant said ***"It really brought me back to school."***. We focussed on the design based learning classes that I observed where the students needed to make an escape room; it has different aspects (hardware, software, storytelling) and the different students had different attitudes (one social; mostly talking to other groups; one student with interest in hardware; one student that sacrificed himself to do the coding and the teacher that was a dealing with other groups as well; so couldn't always dedicate her full attention to one group). We played out three times; with switching trough the roles; and aimed for getting to a result.

Negotiation between students

The scenario was based on an observation in the class; where there was negotiation about who was doing what. This feeling and idea reflected really good in the workshop; the person getting the laptop was 'code responsible' and it was hard to hand over this role; once decided; this person was the designated 'coder'.

Teacher group interaction

The teacher group interaction was also quite representative; during our session of acting out a scenario; we actually got an error code in Arduino that we didn't know or see before; when than asking the teacher about it the teacher is trying the best to solve it. ***"Perhaps you can google it?...Is there anyone in the room who had the same?"***. Which could be perceived as 'weak' by the students one student to another student quietly; ***"He doesn't really know what he is talking about"***.

Intent and concept scaffolding

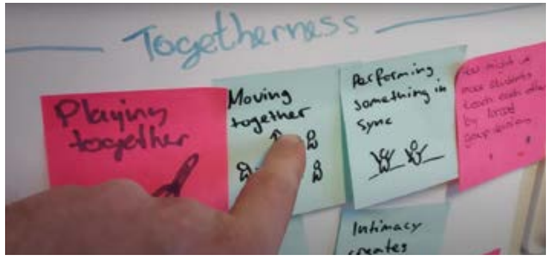
A group of students is 'forced' to define a concept, there might be need for structure to go force the students go through a scenario; first this; than that; than this. So it becomes

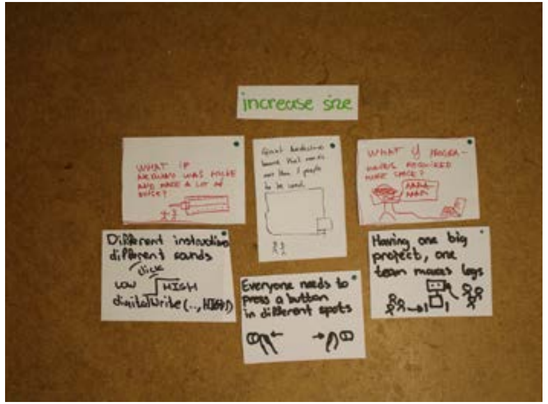


Video showing how the friction is influencing the teaching and learning experience.



Video showing the design based learning process...





"The goal can be really hard, even if we decided that, you have to divide the tasks" - Employee reflecting on the scenario.
In the group interaction it is important to get specific to understand the concept; one participant took this role by asking for clarification: "So you have to have 3 numbers, right?"

Learning and time pressure

The time pressure that is simulated in our acting out of the school scenario shows how peer to peer learning get's lost when the group focusses on the product as result instead of focussing on learning as result *"Can you maybe show me later how you did it?" - Employee acting out as a curious student.*

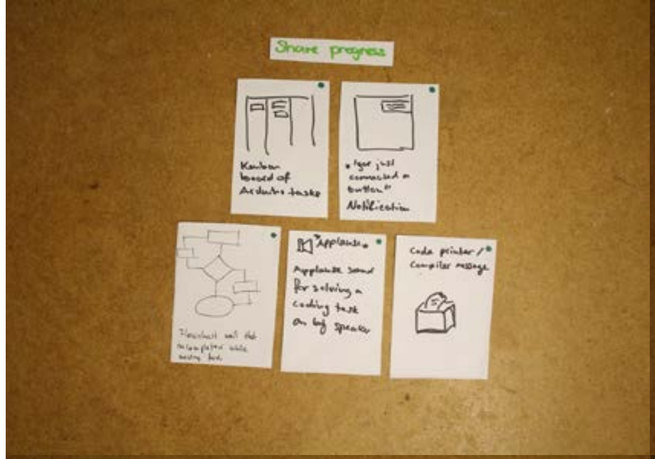
Time planning

Time planning also becomes harder for the teacher and students when they enter unknown territory. *"When you have something pre-defined you can guess how long it would take; but when you come up with something new, all of a sudden that is really different." - Employee reflecting on DBL using Arduino.*

Ideating by using methapors

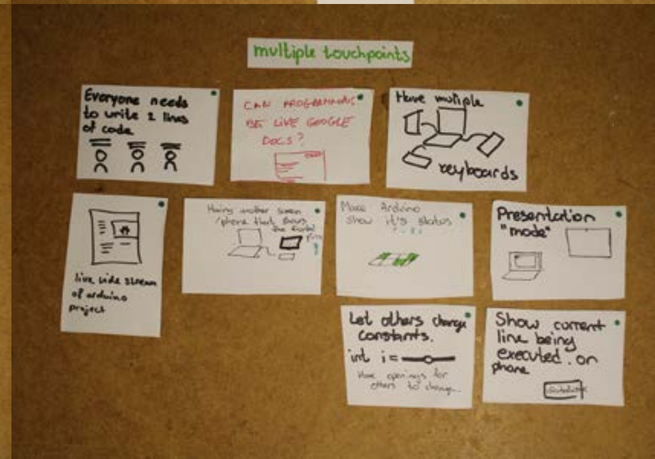
In the last stage of the workshops we ideated on how learning Arduino would be if it was like; learning team sports, learning individual sports, learning an instrument, learning to photograph, learning to dance, gaming, reading or like using social media. These aspects of the methaphors were clustered and shaped into a 'How might we...' phrase that we all did develop 8 concepts around.

We ideated on:
"How might we add more points of feeling that you have achieved something by rewarding them?" and
"How might we create more togetherness by making Arduino more visible and understandable by others?"

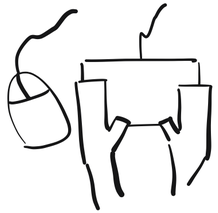


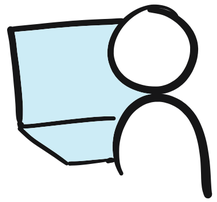


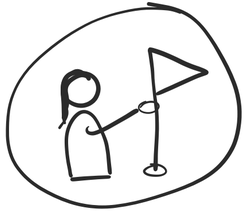












ANALYSIS

Here I wil start showing how the foundations in the results lead to design principles and directions for my final design. Each paragraph starts with the design principle followed by a set of arguments that back up this design principle.

Tools that encourage teamwork

Both in industry and in education teamwork is a crucial part of the learning, this will lead to better skills in authoring and togetherness throughout the project.

The **current tools still focus a lot on individual experiences**, the keyboard and the mouse are allowing one person to interact at the time, besides that the tools (personal computer or laptop) don't encourage passing it around as much as handheld devices. Also the textual representation of behaviour requires individual reading by the user; it doesn't give a overview of what it does in a glimpse.

In the field of CSCW most of the focus is on communication together; although concept from software like google docs is starting to come to the programming realm; in for example the form of Atom teletext. Other collaboration solutions usually involve distributed version-control, such as GIT (known from github), SVN or CVS. Most of these tools focus on remote, advanced and complex projects that many people collaborate on.

Also in the field of programming for education; students (and thus groups of kids) are seen as individual learners that have to achieve certain challenges and reach **milestones as an individual**. This lack of team based tools is due to the individual nature of programming, and might be because the western view on personalised education is reflected through these platforms. You could argue that gamification only focussed on bringing the individual rewards system of games into education; but didn't bring the social part of play into education.

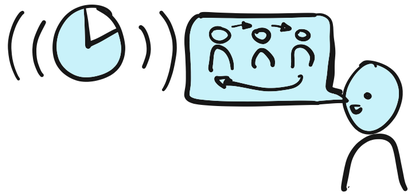
Although these tools are designed for individual use Arduino is **usually taught in groups**. One reason is that otherwise "it wouldn't be manageable" for the teachers. This is due to the friction of the tools, the fragile nature of the components and the fact that there is usually not enough technologically trained support staff to give everyone individual support. Another reason is the lack of tools, partly because of the price, but mostly because of the classroom storage management that it requires. It is easier to keep track of 10 Arduino's, laptops and components than 30 kits. Besides these practical reasons; group work is (as said before) also part of the learning outcomes and the idea is that students are more likely to help each other.

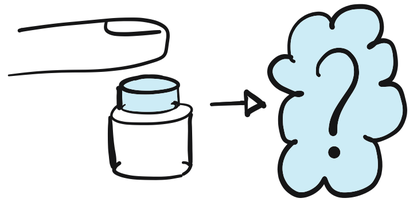
In these groups it is common that **one student is responsible for the 'coding'**, even though teachers emphasise the importance of switching roles in reality this doesn't always happen. One reason is the parallel teaching experience; there is not a clear moment when to switch, since different groups are in different stages of the lesson. So there is no clear moment for the teacher to switch the roles. Switching roles has some friction to it; both on a practical level, moving the laptop in the group or chairs around but mostly on a cognitive level; you have to leave one mindset to move to another one; the person 'owing' the problem needs to leave it and hand-over although according to "Making it stick" this might be a highly beneficial type of friction. Another reason for the 'most' capable student to do the coding is because of time pressure that could be intrinsic or extrinsic; but is visibly present in the current (two)-hourly school scheduling system.

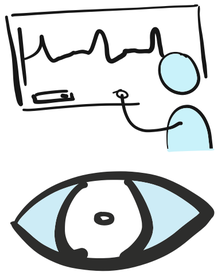
This individual experience of coding is also engrained in the **stereotypes around programming**, both in popular media but also in the job market. This is problematic since this industry is getting more powerful and has to work together with multiple people and disciplines to create solutions that are carried and backed up by more people in the society than only engineers. Also in communication between programmers; authoring is a hugely important part of the job; in the form of hand-overs, writing documentation and tests to make sure colleagues understand what is happening.

Tools that help understanding

"I try to make them switch roles as much as possible; in every exercise they should switch roles; in reality this doesn't really happen, because not everyone is at the same exercise I can't do this with a call for attention." - Technology teacher







Tools can also help to improve this type of communication by visualising the intent and what is happening to increase shared understanding. Currently there is a **lack of feedback** for the students and teachers to understand, troubleshoot and learn. This leads to teachers that are sometimes ‘taking over’ the project from the students to figure out what is working.

“How can I teach students if I don’t grasp what the problem is myself?” - Physics teacher (translated from Dutch)

The debugging tools that should help the students with understanding are **usually introduced after learning the basics** because they are relatively complex to grasp and use, which is problematic. Also these tools require to explicitly define what you want to debug; which makes it hard for beginners since you don’t understand the problem or the procedures to figure out what the problem is.

The tools could also benefit from **more different ways to debug**. Right now debugging is still mostly symbolic visual representations by text, disconnected from the context in which it was created (the code or hardware). By allowing sound to be involved in the debugging it could allow multiple people to participate and it could also become more contextual to what is happening in the hardware.

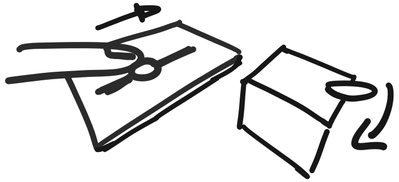
An interesting type of visualisation is the timeline metaphor, several teachers mentioned the power that the Serial plotter gave them in teaching and understanding what is happening. Since Arduino is all about time and how this is related to input and output actions, teachers and students also used this as a mental modal to think and explain what is happening or what should happen.

Also using multiple media channels (video in combination with the timeline) to increase the connected understanding could be a powerful way to let students make connections that would otherwise be invisible to them, especially when getting up to speed with other people’s code, or code from a former session.

Another reason why this is important is because students tend to use code from online sources, imagine if all this code would be documented in such a way that you can explore it; see why things work the way they work; and see the connection to the physical hardware in video, it would probably help a lot with tweaking the code.

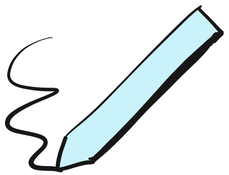
Tools that help interactive exploration

Since programming and code could be defined as a practise and material just like woodworking and wood; we need to make students understand what is possible through (inter)active explorations with this material. Right now several layers of complexity are in between the student and the hardware. What if when they are first encountering a piece of hardware these layers could be broken; so they first learn about the possibilities in an active way; this exploration could be a good foundation for making decisions about what to sketch later on in the process.



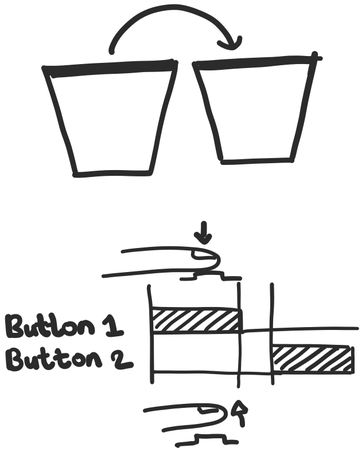
Tools for Sketching

While in the Arduino IDE terms like ‘sketchbook’ and ‘sketch’ are used to explain programmes. It is still hard to sketch different intents in Arduino because of the rigid and sequential nature of the c++ code. In that sense Arduino is still more on the defining side of things than on the exploring side. So what would be a sketch? In design the concept of “Wizard of Oz” is used to “Make the right it, before making it right”. This concept of sketching before developing could help students in group to align since “Defining the goal together can be really hard”. It could also help them to understand the user-centric mindset better; try the concept with actual people before ‘developing’ it. One risk of this approach is that students don’t move from the sketching to developing stage; it could be interesting to explore how there could be a smoother transition where some parts are automated and some parts are faked.



Tools that help contextual understanding

If the students would start with ‘pretotyping’ (Alberto S., 2011) their ideas



by faking the whole scenario they could make a video recording of this and use this as contextual documentation for what they are building. This might also lead to better coupling between iconic, symbolic and enactive understanding. In this way they start learning about the importance of testing and documentation in software development, not as an afterthought but as the basis to start working from. This might also lead to understanding extreme cases better and how to break their own software, which might make them more aware of security consequences of badly tested software.

Instead of building documentation around the code (inline comments); what if the documentation is leading what shape the code has (inline code)?

Tools that focus on transferable learning

Since the field of computer programming is a fast changing field it is important to learn students key concepts that are transferable from computer-architecture to another architecture. These transferable concepts focus more on the capabilities and limitations of the hardware, and essential things like mapping than on the actual limitations of the software. You can already see a trend of non-compiled languages in physical computing (python f.e.) which is already moving towards this more interactive and code-snippet way of working.

Using the timeline methaphor

The timeline metaphor is powerful because of several reasons; it can give you an overview of the action and re-actions between the user and prototype over time. In this way it helps seeing, creating and understanding the behaviour. It can easily connect to video.

Using video

Video is an interesting medium because students in this age range are familiar with various types of video based media platforms (instagram, snapchat, tictoc), it can give an overview of the rich context in which the technology exist (integrated technology). “Video is a remarkably useful tool for analysis” (Yliris, S., & Buur, J., 2007), it captures the interaction between the user and the technology. This combination of the more data oriented way of capturing the user’s actions and system’s reaction in combination with the video could give a good overview of the scenario. Also developments in technology have made it possible to make, stream and record videos for low cost. However there are some risks with video; privacy at schools is a concern, while students use social platforms extensively, in reality parental consent is needed to share and create visual information by them. For this reason the video’s should be processed locally, and might need to avoid having identifiable students. Another risk is that students might see the video more as the end-result than a tool (a sketch); which might distract the students and make the students focus on the quality of the video instead of the quality of their project; it is important for the teachers to highlight that it is ‘just’ a way to document their process and their project.

Why phones

Phones could be a good addition to the current laptop/desktop that connects to the Arduino for several reasons. It is a handheld device which means that students can move around; work on the project from different angles; see the information next to the project. Because most students have smartphones they can all be ‘behind the keyboard’. The technology can nudge them to switch roles easier since everyone has their own device. Since most (if not all) students already have smartphones this solution would be extremely cheap; these devices also have good functionality in terms of video recording and connecting to the internet (and are usually also connected to the school’s wifi) and the students are already familiar with interacting with them. Some risks involved are that students get distracted by their phone; especially in schools where phones in the class are forbidden; they might have less discipline strategies to ignore messages. And there is the risk of students missing out, because they don’t have a smartphone or their smartphone is not up to date. These two concerns are super relevant when developing this concept further; both for the teacher and for the potential development criteria. The popularity of Kahoot (a phone based quizzing app) by both students and teachers indicate that using phones in classroom enviroments is possible.





I’ve been experimenting with top-down projections to explore how this might help contextual understanding, but decided to not go for that direction; for two reasons. 1) price and plausibility of the concept, schools have to invest a lot to get projectors on top of every table. 2) Touchscreens allow for interaction, you can manipulate information on the touch screen quite easily, and the higher resolution and screen just have a better experience.



Code view: By making visible where the code is at a spefic moment it could increase the understanding of what is hapening and why

Board view: By having a board view you could get a better understanding of why the hardware is not responding; you can better understand where the problem might be.



DESIGN

So given all this information; how do we now make a system that would allow for collective, explorative and better understandable types of coding. In my final concept there would be three different stages, in these different stages there is always the need for multiple people to engage; the mobile devices help them to switch roles after each activity. This thesis could be seen as a vision on the future of learning Arduino; but parts of the thesis could be implemented already in ways that would improve the student's experience.

The design quesiton becomes: How might we create tools to enable fluid and social physical computing for design-based learning class students (14-16yr olds) in high-schools.

In the following pages I give you an overview of different sketches, than I will map these different sketches out; reflect on the differences and propose a final result.

Timeline view: By having a representation of the behaviour on a timeline (connected to video) you can see what happened and for what reason.



Arduino Teams
Current paradigm but collaborative

This concept is an exploration how Arduino would look when it would stick to the current paradigm; having excatly the same functions as exist now; which would mean that it would be compatibe with all the libraries and community resources.

The team would be divided into different roles, the 1) function-chooser, the 2) argument maker and the 3) documenter. These roles would switch after an instruction is added to the code.

This prototype doesn't show how naviagtion through the code would be done in this distributed model. This could be a sperate role; or be done by one of the other roles.

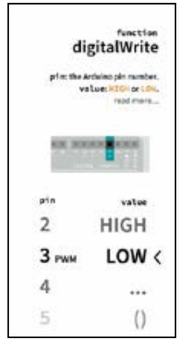




1. Function -chooser

Endless scroll for endless knowledge

This interface encourages exploring; by using the scrolling interaction in the bottom in combination with short texts and the possibility to read more the students curiousity is triggered.

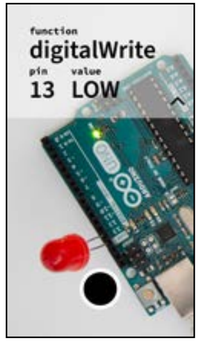


2. Argument makeр

Arguments that make sense

This interface for the 'argumenter' is made to make the user understand what the options are; and how they relate to the real world.

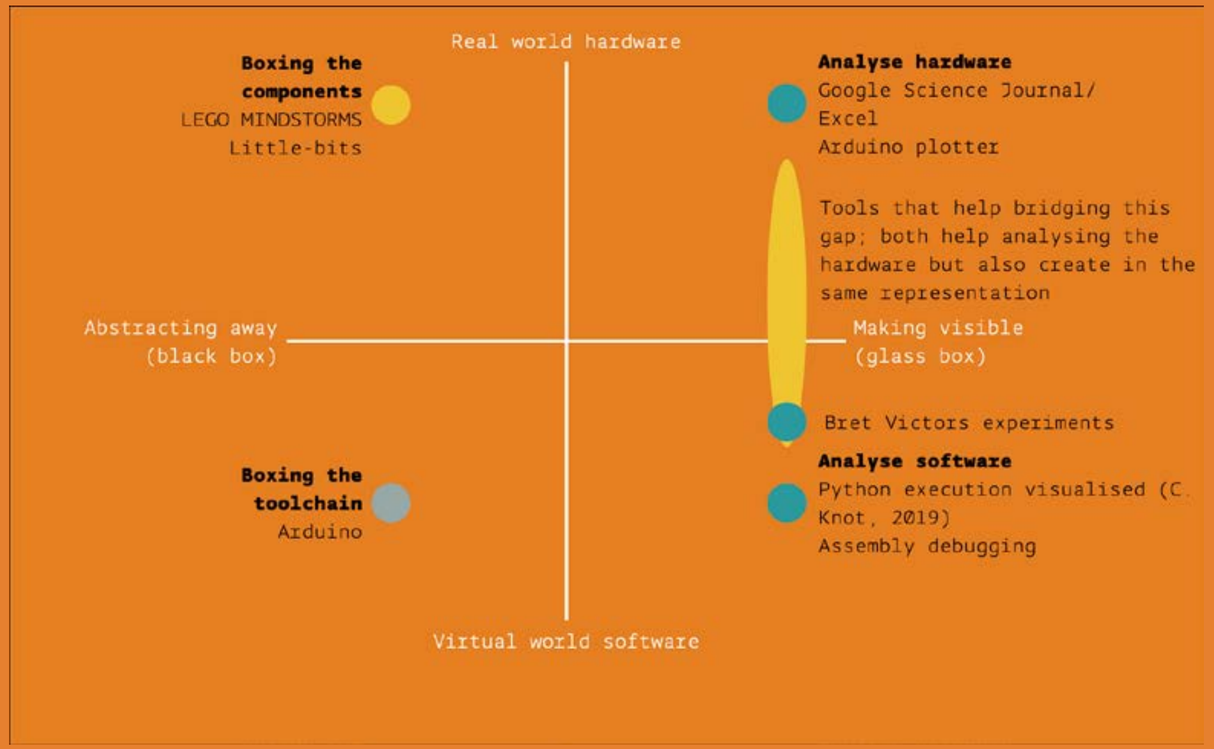
The values should immidiately be sent to the output ports of the Arduino.

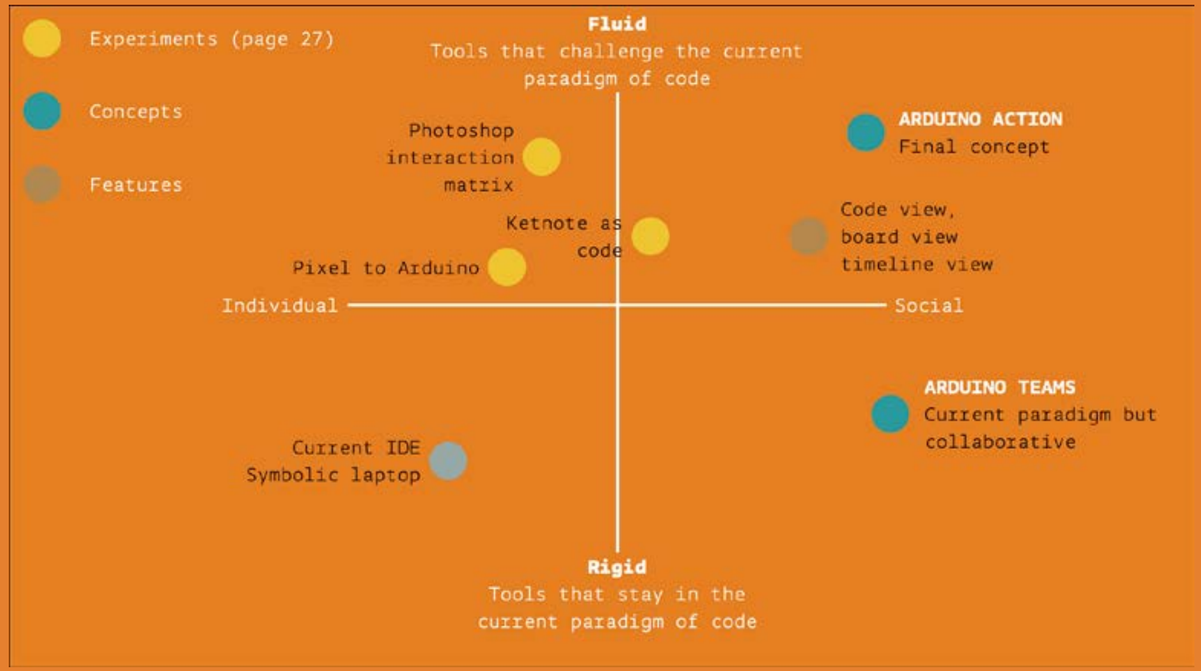


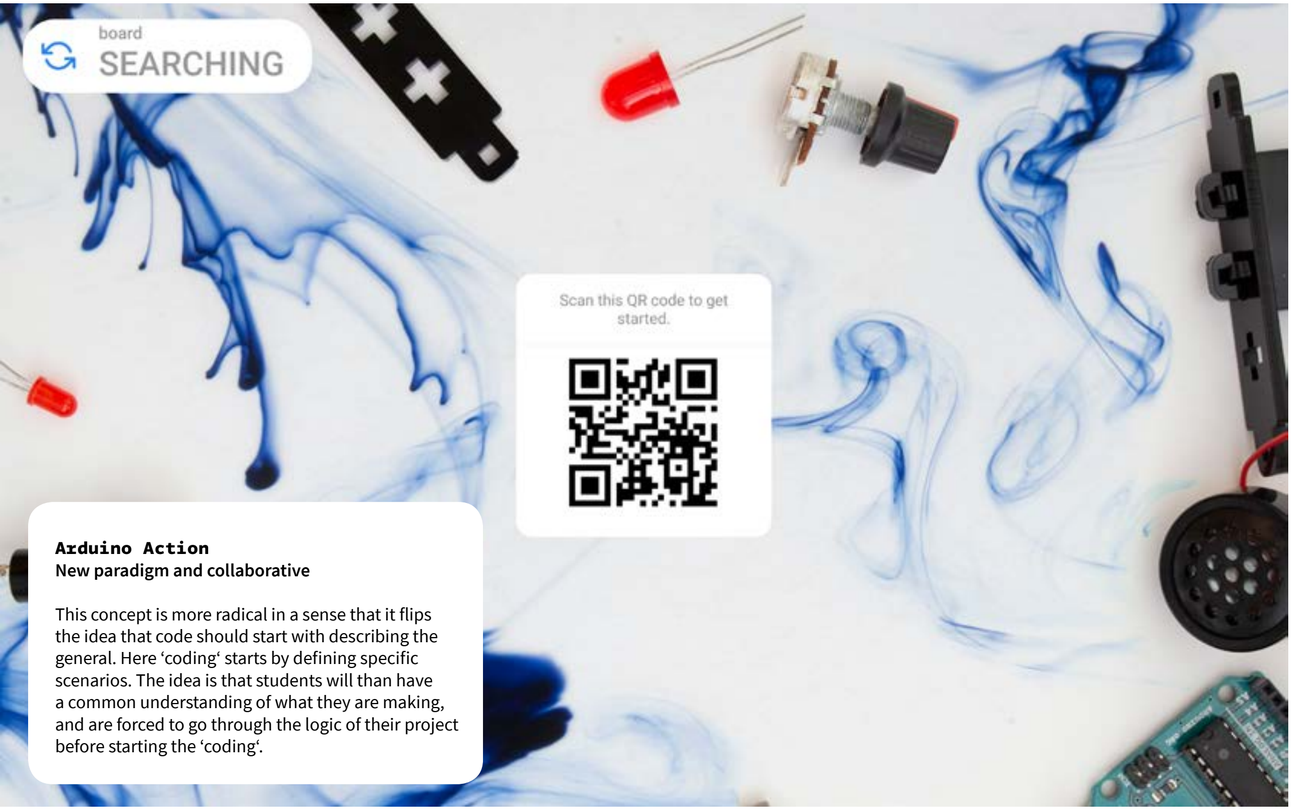
3. Documenter

Real life documentation

Instead of writing documentation documentation can be created by pictures. Students can use text or emoji's on top of the images to make abstract 'invisibe' behaviour documented.


Real world hardware
Boxing the components
LEGO MINDSTORMS
Little-bits
Analyse hardware
Google Science Journal/
Excel
Arduino plotter
Tools that help bridging this gap; both help analysing the hardware but also create in the same representation
Abstracting away
(black box)
Making visible
(glass box)
Bret Victors experiments
Boxing the toolchain
Arduino
Analyse software
Python execution visualised (C. Knot, 2019)
Assembly debugging
Virtual world software


Experiments (page 27)
Concepts
Features
Fluid
Tools that challenge the current paradigm of code
Photoshop interaction matrix
ARDUINO ACTION
Final concept
Ketnote as code
Code view,
board view
timeline view
Pixel to Arduino
Individual
Social
ARDUINO TEAMS
Current paradigm but collaborative
Current IDE
Symbolic laptop
Rigid
Tools that stay in the current paradigm of code



Arduino Action
New paradigm and collaborative

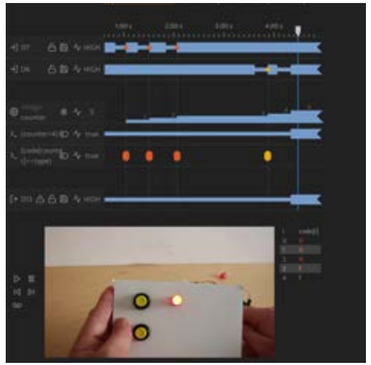
This concept is more radical in a sense that it flips the idea that code should start with describing the general. Here ‘coding‘ starts by defining specific scenarios. The idea is that students will than have a common understanding of what they are making, and are forced to go through the logic of their project before starting the ‘coding‘.

Visual style

The visual style should be clear, don’t look to scary but at the same time shouldn’t be to childish. Since the concept is already challenging the status quo you could argue that the visual style should follow this. However to make users understand what is happning it is good to stick to current paradigms in visual design; also it will put more emphasis on the concept; if I would go for a more radical style of UI than people will be distracted by the way it looks; which wouldn’t support the concept. The information shown on the personal screens is also represented on the bigger laptop or desktop; in this way the phones are like a ‘smart keyboard’ that helps the students in understanding



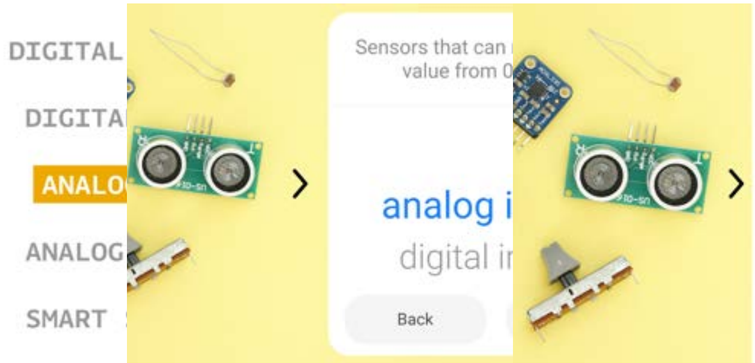
Colorfull but content makes it look simple



Adobe style type of exploration



Arduino branded style



Google material UI

Scrolling UI in some typographic format like in Arduino teams

Encourage exploration by design

Make exploring easy, fun and encouraged by the tools, give hints to more documentation/ concepts.

Rich representations

By combining text, visuals, contextual and the enactive representations students get to make connections that were before abstract. Students ‘know‘ what digital inputs are; but wouldn’t call it that name; since this is Physcial computing specific jargon. By seeing what different digital inputs exist they start creating a mental model around it.

Document your explorations

The explorations can be documented and used in later parts of the project.

Explore (Getting to know the material)

In this stage the students will explore and document what specific components can do; this is to get familiar with building the electronics, testing if everything is connected and making re-useable documentation for their later stages. The goal is to get them inspired and create basic understanding of the capabilities and limitations.
In this stage they test both the sensors and actuators; sensors by recoding how the values change based on human input; actuators by changing the ‘commands’ to these by controlling the actuator on a phone to send different values to the actuator. This is also the moment where important ‘snapshots’ could be taken; see these as important values that relate to the sensor or actuator.





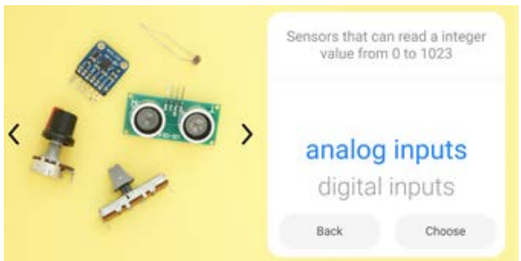




The alarm clock scrolling UI is used because it is an enjoyable experience, thus encourages exploration by design.

Selecting/browsing the types of componnents

Here the student will be encourages to explore the different types of components; by allowing the student to scroll through the various types of components; the arrows on the side will lead you to specific components and how they work.



Friction

The order of this categories is random and switches every class; so that students don’t learn the position; but the concept.

Demonstration videos

There will be boomerang type of videos that show the concept of the category by demonstrating an example. Values will be represented in various ways to enhance the understanding. Teachers and students could also make their own explanations and use this in presenations. By combining, textual, video and visual representation the aim is that students get the underlaying concept and not just the representation.

Friciton

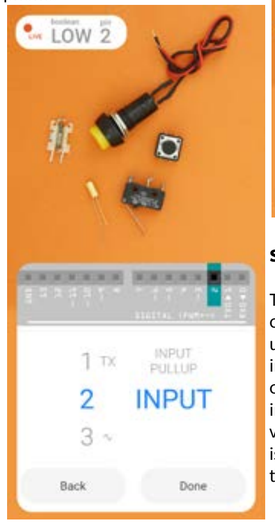
A random representation is shown; when the user presses the box it switches to another one.





You're live

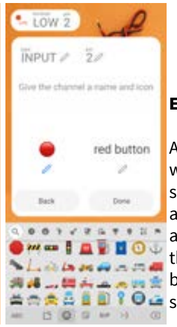
You can now see the actual value read by the Arduino on your phone.



Understand why

The students will imidately see somethign weird; the value is constantly flickering between LOW and HIGH. Instead of explaining the need for pulling resistors, they can understand it by exploring it.





Show complexity

The main goal is to show the complexity, so it becomes more understandable; here the pin and input type is selected; and you can immidiately see how that influences the live value. Also a visual representation of the board is shown; which might help finding the pin.





Emoji's in code

A set of emoji's are an excellent way to communicate what a sensor/actuator is. They are using a small space, are visual distinct and could be more descriptive than long wording. There might be need for more Arduino/coding specific emoji's in the future.

Sonified logic

When tapping on the button on top they can change the representation; including audio; so the group can hear when it is working.

Shared understanding

When the channel is being created the current value is already broadcasted on the big screen; which helps the group understand what is happning.

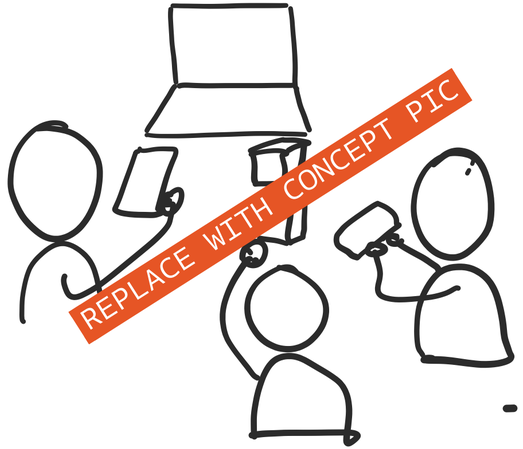


Output control

When the channel is an output channel the student can control this channel by using the phone. This helps understanding the field of action of this component. Students can explore different types of control.

Record

When they know the capabilities of the sensors and actuators, they can start putting these sensors and actuators in a scenario; by recording this scenario they are forced to think through all the steps; and they have a common understanding of what it is they are making. By recording cases of people acting weird; they can learn about the importance of testing.



Story first, coding later

The students are encouraged to communicate the story first; this helps the group having a common understanding of what they are aiming for. It helps the teacher understand the different steps, and it serves as a back-up prototype.

Testing

This approach to coding also helps students to think about testing; how could you 'trick' the system; which might also help them understand things like 'hacking' and security issues with technology.

Snippets

The structure of the code has the possibility to use blocks to influence one channel by another channel. In this way you can combine ‘faking‘ with coding and itterate fast.

Some of the functions that are included in these snippets include

Map
Re-map the value from one channel to another channel

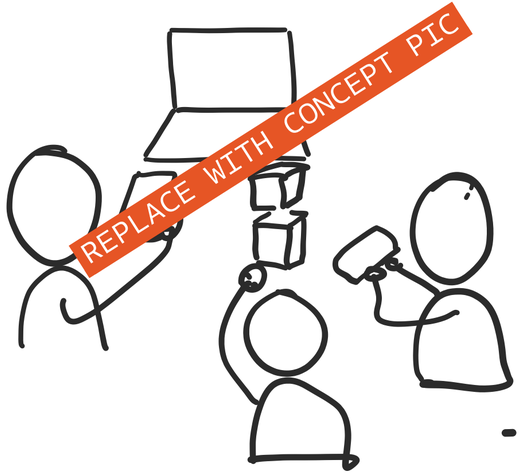
Event
Triggers an action based on a rise or fall event from a binairy channel

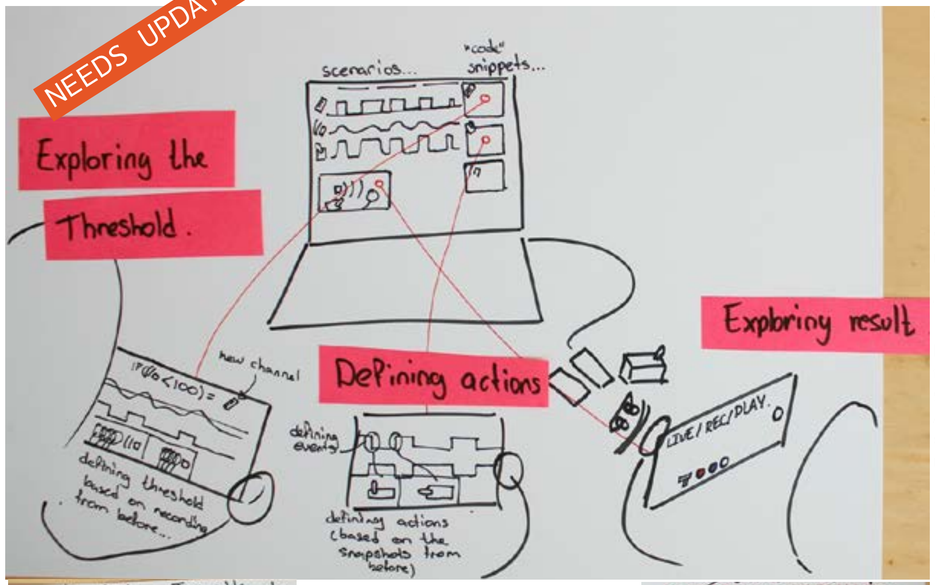
Threshold
Sets threshold for a analog or integer value

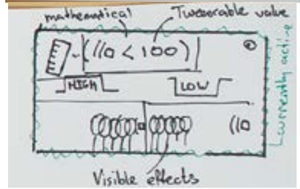


Generalising

This is the part where the students make the actual ‘code’, they do this by each defining a snippet of code; this might connect one channel to another channel; or do some mathematical manipulation on a certain value. They should understand how their changes impact the result; and how it matches; or doesn’t match the test results.

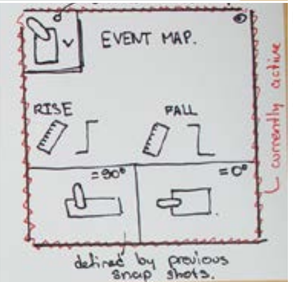






Tweaking threshold

By using previously recorded documentation from the exploration they can use this to define thresholds.



Event

The event function is something students can use to define when something should happen. Again they can use previously recorded documentation

Generalising

In this phase students work together because each student is working on a ‘snippet‘ on their phones. In this way they learn to work with causalities and they can also work with semi-automated scenarios. Where one input channel is still influenced by the students app; while the others might be responding to this value.

Timeline

In the timeline they can see how their ‘code‘ would infleunce past scenarios in the form of an overlay. Also new recordings and live explorations can be made to get more enactive understanding.

FREQUENTLY ASKED QUESTIONS

Here I will try to answer hypothetical questions from different stakeholders, some I might be able to answer better than others.

Questions from a teacher

How will this change my teaching practise?
Good question; first of all, we need more trials to figure out what would be the best way to fit this into your current curriculum. The goals of this project is to support more collaborative and active ways of learning, and use different representations to give the students an understanding of the learning content as soon as possible. The goal is that you as a teacher can focus more on supporting active learning instead of being technical support in the classroom. My assumption is that because students (and you as well) see better what is happening they can start exploring easier. And when they have problems you can see easier what the problem is and ask the right questions to scaffold their understanding.

What do I need to know?
Depending on your own experience with using digital media in the classroom, you might need to understand how to make clear rules about using phones in the classroom; what is allowed? These boundaries need to be clear from the start, especially since these tools also allow video and audio recording to do documentation; which if used the wrong way might violate the laws and regulations of your school.

Give students time to explore; time pressure can help students to package their concept, but it can also lead to learning-avoidance (the students that already understand it, will do the job). Let them explore the components; but give clear guidance on the fact that they need to document their explorations.

You could for example let them explore; but they have to show three different uses of the sensor or actuator that the group is exploring. This will help them creatively understand how to use sensors or actuators; which will help them in later stages. At the end of the class they present their explorations to the class; so they inspire each other with possibilities.

You as a teacher know best how to divide group work and work with roles in the process. Since the parallel learning experience makes it hard to force students to switch roles you can use this tool as a way to enable this switching of roles. For the students this might seem hard in the beginning (since they might be engaged in a specific part of the project), but it will help them with understanding the whole and it will lead to valuable hand-overs between the students.

How do I start?
The best way to start is by exploring the tool yourself, try to make some material in the documentation tool that might help student's understanding. By bringing it close to the context of the students (think about the school, environment around the school) you can help them making connections from this design based learning class to the real world.

Questions from students

Will I be prepared to start a computer science degree after using this tool?
You will gain skills attitude and knowledge that are an important part of what programming is in the real world. Perhaps you will be differently prepared than other students that learned a programming language like python. But some of the concepts like documentation, testing and how to work together with other students are really important for a professional programmer, which they might not have learned that much. If you want to focus more on how computers work under the hood; you might want to learn some low level programming language like C, this will teach you more on how the bits and bytes inside the chip are moved around registers. But since you already have a basic understanding of what binary values are and how they are used in the world of physical computing this will help you to understand this. Also look out for cool hackathons, makeathons or places where you can learn with others; it will help you to see what you like to do in the future.

Can I share my experiments on instagram or tictoc?

Great that you want to share you're experience with others, it might inspire others and make them understand these topics as well. You have to ask your parents, your teacher and any person that is visible in these experiments; your school might not allow you to share things that you explored in the classroom. But if your parents are fine with you sharing this; you might be able to buy an Arduino some hardware and make these videos at home.

Questions from parents

I don't want my child to bring a smartphone to the school, can she still join?
Of course, your daughter is definitely welcome in the class. From my own experience as a teacher I have sometimes solved this by letting students use my phone, you should ask the teacher what the possibilities are. It is also the schools role to give the students the right learning material, so perhaps they might need to buy some smartphones for these occasions. Also I fully understand you're concern regards smartphones and students; but be aware that they have to learn to use it responsibly. By not allowing it, you might not develop this skill of using digital and social technology responsibly.

Questions from a technologist perspective

I've been working as technician as several schools, and the wifi is usually sh*t, how do you think this will work?
The main scope of this thesis is to sketch a future vision on DBL education. For this I allowed myself to not focus too much on the details that would block this vision. I believe that if we can shoot rockets to Mars, we should be able to connect a phone to a laptop in a reliable manner; trough bluetooth, wifi or some other form of communication that we don't know yet, perhaps this might be good for you as a technician to have an argument to invest in the wifi networks at schools.

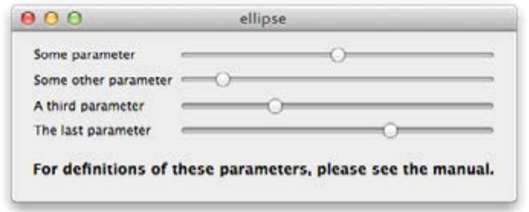
Questions from Bret Victor

These questions come from the learnable programming essay on Bret's website.

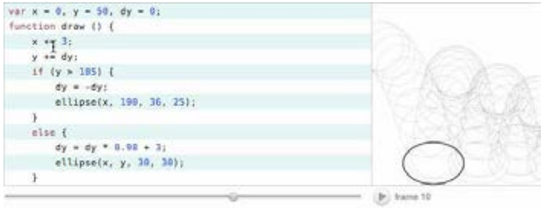
What do these words mean?
The meaning of the words is made visible, tangible and understandable by almost every way possible. Students for example learn what HIGH means by seeing it represented in various ways; using sound, using video of an output, using video of an input, in the context of the learning material and in other contexts for example the button of a blender in the kitchen. A next step of making students understand these words is probably by putting them in AR in the environment that they are currently in. Like in the Vuforia Spatial toolbox or MIT's Spatial Object programming. Although AR is usually not the most collaborative medium, since is still has the same appearance as sunglasses. I dind't focus that much on AR or top-down projection; since it is also possible with current existing technology; and it would shift the focus of the concept from the core; making things understandable and collaborative to the core being using AR/ table projections.

What happens when?
The timeline metaphor is one of the foundations of the current concept; it means that students literally can see what will happen when; and if needed scroll back and forward in time. However they can't look into the future; except when the future is following an exact path that is defined in the past; which I'm not sure if we can call that an actual future.

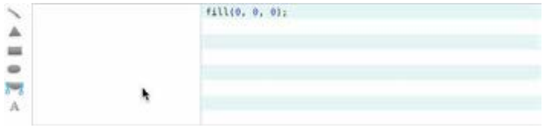
What is the computer thinking?
Again the channel-timeline metaphor makes everything visible from the variables used to the events that are triggered. In this way students can see at each moment (and in the past) what the computer was thinking. Besides this timeline; students can also see what the computer is seeing through the channel controller on their phone.



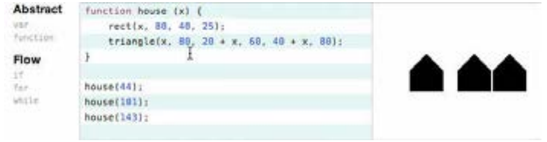
(B. Victor, 2012) This interface reflects how current programming languages would be translated to UI. Altough this is already an improvement since it limits to inputs to numeric values.



(B. Victor, 2012) This interface shows how time can be made visible for the programmer to increase how changes in the code have impact on the whole.



(B. Victor, 2012) This interface shows how students can pick visual representations of shapes and then use this to code; why descibe an ellipse by the word ellipse and not by the shape of the ellipse.



(B. Victor, 2012) This interface shows how students start with a concrete shape a house, but than abstract more and more of this house, in this video f.e. the height of the house.



The Google science kit shows that it is possible to have one party develop the software and another party selling the hardware. Altough there should be clear rules who is responsible for what. Since the first version of this kit by Sparkfun is discontiued.

Can students start somewhere and then sculpt?
Yes students start somewhere; by exploring the possibilities of the hardware and then they can record this to use this as a starting point to sculpt upon. Also by recording they can reflect in the group and record another iteration.

Do students start concrete and then generalise?
Yes exactly, students start with super specific; concrete recordings of scenario's that they act out in a group. Afterwards they generalise by mapping the channels and creating automated behaviour; because they can choose to automate parts and act out parts of the experience, students can slowly move from concrete to half-generalised to fully generalised and test this generalisation on past and future experiences.

Questions from the business?

All nice and well, how is this fitting into a sustainable business model?
Good question, since having this land in a good durable and fair model is important. I think there is a lot of interesting possibilities around the open-source community, currently Arduino is also having a model where the IDE is open-source and free to use; the hardware and curriculums are available as educational package. Although it would also be interesting to involve the community into making and sharing interesting ideas around the curriculum (also because these tools allow students and teachers to document in interesting and engaging ways). We could also sell the software as a service; but give the source code for free (so people can run it locally, with still an enjoyable installation experience) but if they want to use our server infrastructure they have to pay a yearly fee for each subscription. It is important that there is a responsible entity doing maintenance and updates so schools and teachers can rely on it being available in the future. Another option would be this would be sponsored and developed by an external party like the Microsoft Makecode or Google science journal and the Arduino science kit since it is a software-only solution the initial investment will be relatively high, but the cost for maintenance (especially if the community is helping there) is really low.

Questions from Stoffel, programme director for Interaction Design

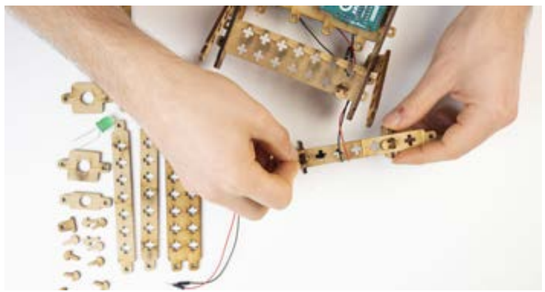
Why didn't you focus on the hardware?
The simple answer would be, that it is not in the scope of this thesis; but I'll describe why I think that it is not. Although I do have a personal interest for making physical things, tangible interfaces and rich interactions that go beyond the screen.

The cost of making and customising hardware is different from software. Cost in software is defined by how far it is from the current paradigm; everything that is close to the current paradigms is cheap. One example of this is blockly; since Google made an open-source library to make your own block based programming environments; it allowed start-ups and bigger companies to implement this type of coding without big costs, this is not only due to development costs but also due to branding, design and research costs that come with developing something new. So software tends to 'cage' itself. With hardware this is different cost is definitely also defined by the volume; but mostly by the materials that go into it. Both of these are reasons to stick with off the shelf DIY components since the volume is big and you don't have to manufacture a box around it.

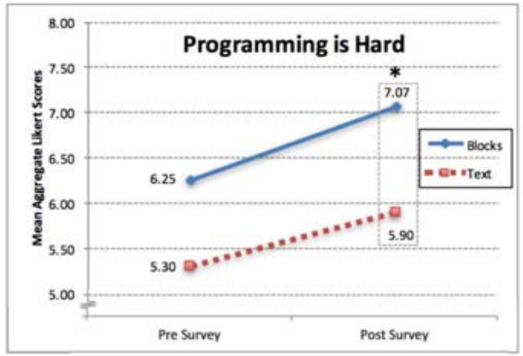
With cost I don't mean only the price in a currency but also the cost on the environment, cost for shipping or cost as in developer hours spent on making something. These unfortunately don't always match the cost in currency.

A personal reason to not focus that much on the hardware side of things is that in my internship at Arduino, I've been looking into the hardware; in Arduino mostly on how to make holders for components that don't hide their complexity; but still allow for more flexibility and creativity by the students, so project can be beyond the breadboard.

By making students interact with DIY hardware instead of ready made products they start seeing that what they are using in their daily life is the same as what they are making in the class. When you unscrew a piece of kitchen equipment you would find similar components to the once you are



Initial design of a system for allowing freeform building with DIY components.



(D. Weintrop, 2017) This chart shows that both students that followed the text based programming and the block based programming had the same change of percpetion of the fact that 'programming is hard', not corresponding to the other confidence factors measures. The researchers assumed that this was due to the fact that students didn't see what they did with the blocks as 'programming'.

using in the class. Of course even DIY hardware is designed to be easy to handle and is usually bigger and more accessible than the SMD components in your smartphone; but the basic concepts are the same.

So you did focus on 'real' hardware why didn't you focus on 'real' programming?
I think the software I'm proposing is as real as other types of programming although in the paper on block based programming the students tend to define the learning experience in the block based programming language not as 'programming'. The same might be true for end user programming environments as Excel. This shows how boxed our idea of programming is; if we make it better, more accessible, more understandable or more visual it suddenly doesn't look like programming anymore. This might be a reason to not use the term programming, so this image of how it looks like today is not attached to the project. But that would also reinforce this idea that programming is only written text and not the concepts behind it.

What are the aesthetics of interaction?
I started of my project in changing the aesthetics of programming and interacting with physical computing; although using the term 'aesthetics of interaction' is confusing and thus blocking people not from 'our' background in interaction design. I used metaphors and the word feeling more than aesthetics; since it covers the load. 'How would programming be if it feels like doing a team sport?'. Using the painters metaphor was also a nice way; since there is a clear distinction between the painting as an art and painting as in whiting a wall.

That was not the question, what is the aesthetics that you're aiming for?
Ow, I hope it was clear from the report; the goal is to create more togetherness and teamwork while still being partly individual responsible since research shows that this is a great way to learn. The goal is to put friction in the right spots, and take friction away from the spots where it decreases understanding. Friction in the right spots are to decide on a specific scenario as a group, which can be hard but by putting it out there in the beginning it opens up the discussion at the start instead of at the end. Putting friction in changing roles, it is hard when you're getting used to a specific role to switch, but this switching will force students to help each other, it will bridge the understanding from the different angles and create empathy for the different roles. Friction should be taken away from parts where the friction is 'blocking' imagination or exploration. When first encountering a sensor or actuator the best way to learn what it is doing is by playing with it, not by having to write 10 lines of code to make it do or read something.

What are you most proud of?
I think this idea of collaboratively working on programming (or defining behaviour) in education is quite a novel idea that could have tremendous impact; since collaborative active learning seems to work great according to research and has the side-effects that are harder to measure but perhaps more important, such as interpersonal-skills, empathy and the understanding the value of communication. It would be interesting to see these concepts getting into platforms like code.org.

Programming is like a canvas that is too big to paint as an individual; so why do we still teach it as an individual sport?

REFLECTION

In this thesis I learned a lot from the field study doing the workshop and reading materials related to learning and teaching. The field study was crucial for verifying and exploring the space; it gave me embodied understanding of what is happening in the classrooms which helped me and gave me confidence to move forward with my concept.

There has been a lot of parts of this projects that move into highly complex and tricky fields like the programming paradigms and how they define our perception of what code is. The benefit of being a designer is that you can navigate this space by making; not by knowing everything that is said about these topics.

What surprised me is how effective even small active collaborative learning interventions in frontal education have and knowing that we have this information from the 80's it is surprising that we still form our classrooms as we did before.

Another surprise is how we as a society accepted certain norms and stereotypes of programming over time. In the begin of human computer interaction it seems like our imagination was wilder and sometimes I feel like progression of technology has made us more conservative in views on different futures. We as a society are becoming old and grumpy, resistant to change; while at the same time on the surface level programming languages, frameworks and tools change rapidly there seems to be a lack of deeper experiments.

What would I do differently?

In my original planning there was more time and iteration assigned to doing user-tests in the field to verify certain steps in the design process. Due to circumstances of the current COVID-19 situation it was harder to reach teachers and students, even remotely. Now I did some of this by having remote workshops with educational experts and teachers; which helped me gain confidence and getting feedback on specific details or important parts of the projects. But what was lacking is user-validation with students in a classroom context; which is obviously necessary when this would become a product.

How would I approach the problem space now?

I think my original intentions were quite close to my final result; when approaching the problem space now it might help to record screens from students while working on their Arduino projects; in this way their privacy might be preserved while it could still give me an overview on how all the groups were doing. Having in some way recordings of the actual procedure of troubleshooting and sense-making besides my own notes and memory would help to tell the story and it would be easier to map opportunity areas based on frequency.`

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APPENDIX

Toekomst van Arduino onderwijs - Future of learning Arduino
Deelnameformulier / Informed consent form

Geert Roumen
Umeå Institute of Design
gero0009@student.umu.se

Deelnemen / Taking part

NL1	Het doel van het onderzoek is mij verteld, ik begrijp dat dit onderzoek doelt op het uitbreiden van wetenschappelijke kennis.	☐
EN1	The **purpose and details of this study** have been explained to me. I understand that this study is designed to further scientific knowledge.	
NL2	Ik heb het informatie-blad en dit toestemmings-formulier gelezen.	☐
EN2	I have **read and understood** the information sheet and this consent form.	
NL3	Ik heb de mogelijkheid gehad om vragen te stellen over mijn deelname.	☐
EN3	I have had an opportunity to **ask questions** about my participation.	
NL4	Ik begrijp dat ik vrijwillig deelneem aan dit onderzoek en altijd het recht heb om de deelname te beindigen zonder dat ik daarvoor de reden moet geven.	☐
EN4	I understand that I am under no obligation to take part in the study, have the **right to withdraw** from this study at any stage for any reason, and will not be required to explain my reasons for withdrawing.	
NL5	Ik geef aan om deel te willen nemen aan deze studie	☐
EN5	I **agree to take part** in this study.	
NL6	(optioneel) Ik ga ermee akkoord om opgenomen te worden (audio/video) als documentatie.	☐
EN6	(optional) I'm agreeing to be recorded **(audio or video)** as documentation.	

Gebruik van informatie / Use of information

NL7	Ik begrijp dat alle persoonlijke informatie die ik deel zal worden geheimgehouden en dat dit anoniem gebruikt zal worden door de onderzoeker(s).	☐
EN7	I understand that all the personal information I provide will be treated in **strict confidence** and will be kept anonymous and confidential to the researcher(s).	
NL8	Ik begrijp dat relevante geanonimiseerde quotes gebruikt kunnen worden in publicaties, verslagen, web-pagina's en andere onderzoeks resultaten.	☐
EN8	I understand that **anonymised quotes** may be used in publications, reports, web pages, and other research outputs.	

Name of participant	Signature	Date
Geert Jacob Roumen		
Name of researcher	Signature	Date

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Toekomst van Arduino onderwijs
Participant Information Sheet

We would like to invite you to take part in our study. Before you decide we would like you to understand why the research is being done and what it would involve for you. One of our team will go through the information sheet with you and answer any questions you have. Talk to others about the study before making a decision if you wish.

What is the purpose of the study?
The purpose of the study is to observe how tools like Arduino are used at the moment in education, to develop a new way to interact with these tools so they fit the curriculum better and/or enable other possibilities and concepts. Finally it will lead to an interactive prototype that communicates the cange in experience.

Who is doing this research and why?
Geert Roumen has chosen to focus his Master thesis in Interaction Design on this topic because it triggers his interest and is relevant (because the next generation needs to understand and control technology).

What will I be asked to do?
This is dependent on the context in which the research happens and what has been agreed on between you and the researcher.

Observation:
You will be asked to allow the researcher in your context (classroom). The researcher will observe what is happening in the classroom and make notes on to memorize the dynamics. This might be followed by a co-reflection session (see next paragraph).

Co-reflection:
The researcher will reflect together with you on the lesson that happened to understand and document the context (without intruding the privacy of the students). This could be done in the classroom to enable the rich information and memories that exist in context.

Interview/Workshop:
The researcher will try to understand your current practises from a higher level; understand you're approach, learning goals, ambitions. To do this there might be several methods to make it more engaging, collaborative.

Once I take part, can I change my mind?
Yes. After you have read this information and asked any questions you may have if you are happy to participate we will ask you to complete an Informed Consent Form, however if at any time, before, during or after the sessions you wish to withdraw from the study please just contact the main investigator. You can withdraw at any time, for any reason and you will not be asked to explain your reasons for withdrawing.

However, once the results of the study are aggregated (expected to be by 5th of June 2020), it will not be possible to withdraw your individual data from the research.